

GEN-3 SAFE HELICOPTER

DOCUMENT 00

10-TON / 7-ROTOR SAFETY-CENTERED BASELINE

Fixed Safety Capsule Configuration

Seven 6.0 m Rotors • Four Turboshaft Prime Movers

Conceptual Design Document • Revision 00-T10-R6-Full

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PROPRIETARY CONCEPT INFORMATION

Concept / Pre-Feasibility Baseline — Not Certified

Baseline lock: The earlier 100–200 ton concepts are superseded for this product baseline. This document establishes the new compact 10-ton architecture with seven **6.0 m** rotors, **1,700 L** fuel capacity, four turboshaft-class prime movers (installed power target 2.5–2.8 MW), and a fixed non-jettisonable safety capsule protected by a ten-layer safety architecture.

1. Document Control

Item	Controlled Value
Document Number	00
Revision	00-T10-R6-Full
Project	Gen-3 Safe Helicopter
Product Baseline	10 metric tonnes / 7-rotor compact platform
Rotor Configuration	7 rotors, nominal diameter 6.0 m each
Total Disk Area	197.9 m² (geometric)
Capsule Configuration	Fixed, non-jettisonable safety capsule
Fuel Planning Capacity	1,700 L nominal
Prime Movers	4 × turboshaft-class (650–700 kW class each)
Installed Power Target	2.5 – 2.8 MW
Overall Envelope	17 m × 17 m × 4 m target envelope
Occupant Capacity	7 (1 pilot + 6 passengers)
Founder	Amirbahador Atai
Status	Concept / pre-feasibility; engineering validation pending

This document is a design baseline, not a type certificate, flight manual, structural substantiation package, or production drawing set. Quantities identified as targets, planning values, or preliminary estimates must be closed through analysis, supplier data, component testing, system integration testing, and the applicable certification process.

2. Executive Summary

The project is deliberately shifted from the previous very-large conceptual vehicle family to a compact 10-ton rotorcraft intended to be easier to prototype, demonstrate, finance, and progressively validate. The defining product feature remains occupant survival through a dedicated ten-layer safety architecture, expressed in a manufacturable fixed-capsule configuration.

- 10,000 kg maximum design mass target.
- Seven primary lift rotors, each with nominal **6.0 m** diameter (total geometric disk area $\approx 197.9 \text{ m}^2$, disk loading $\approx 50.5 \text{ kg/m}^2$ at 10 t).
- **1,700 L** nominal fuel planning capacity ($\sim 1,360 \text{ kg}$ at 0.80 kg/L Jet-A).
- $17 \text{ m} \times 17 \text{ m} \times 4 \text{ m}$ target overall envelope.
- Central fixed safety capsule with thick multi-angle carbon-fiber roll-cage, high-stroke mechanical isolation, soft damper-mounted seats, and 6 mechanical damper legs.
- Seven main mechanical damper landing legs with thick soft plastic tip sections.
- Four turboshaft-class prime movers (650–700 kW class each) feeding a segmented electrical distribution system to seven independent rotor drive channels.
- Installed power target **2.5–2.8 MW** (providing margin over the $\sim 1.94 \text{ MW}$ approximate real hover power at 10 t).
- Aircraft-level emergency recovery concept with 8 advanced parachutes.
- Fire detection, isolation and suppression as independent safety functions around critical zones.
- Architecture intentionally modular so later versions may introduce a detachable capsule without requiring the entire platform to be redesigned.

This full revision closes the propulsion architecture direction, updates fuel capacity and performance estimates, and retains complete conceptual honesty regarding the status of all numbers.

3. Frozen / Controlled Product Baseline

Parameter	Baseline	Status
Maximum design mass	10,000 kg target	Controlled target
Rotor count	7	Controlled
Rotor diameter	6.0 m nominal	Controlled
Total rotor disk area	197.9 m² theoretical geometric area	Calculated
Disk loading at 10 t	~50.5 kg/m²	Calculated
Disk loading at 15 t lift case	~75.8 kg/m²	Calculated
Fuel capacity	1,700 L nominal planning capacity	Controlled target
Fuel mass (reference)	~1,360 kg at 0.80 kg/L	Planning value
Prime movers	4 × turboshaft-class units	Controlled direction
Power class per unit	650–700 kW class	Target class
Installed power target	2.5 – 2.8 MW	Controlled target
Overall envelope	17 m × 17 m × 4 m	Controlled target
Safety capsule	Fixed / non-jettisonable	Controlled
Capsule isolation	Mechanical isolation + multiple damped load paths	Architecture
Primary safety philosophy	Ten-layer containment, isolation, energy absorption, fault tolerance, emergency recovery	Controlled

4. Vehicle Architecture

The 10-ton platform is organized around a central safety capsule and a distributed lift system. The seven-rotor arrangement is treated as a system-of-systems rather than seven isolated propulsion units: each rotor station has mechanical, electrical, thermal, control, and structural interfaces.

Four turboshaft-class prime movers feed a segmented high-voltage electrical distribution system that supplies seven independent rotor drive channels. This architecture allows any single prime mover or any single rotor channel to be isolated without immediate total loss of lift authority.

- Central occupant capsule positioned close to the vehicle center of gravity.
- Primary airframe forms a load-carrying perimeter around the capsule.
- Rotor stations are distributed around the vehicle envelope to provide lift redundancy and control authority.
- Fuel and hot machinery are separated from the occupied capsule by dedicated containment and isolation zones.
- Landing and ground-impact loads are routed through sacrificial or energy-absorbing structures before reaching the capsule.
- The architecture prioritizes controlled graceful degradation rather than reliance on one heroic emergency device.

The compact architecture is intended to reduce development complexity relative to the superseded large concepts while retaining the distinctive safety-centered identity of the project.

5. Ten-Layer Safety Architecture

The ten-layer architecture is the core identity of the aircraft. Layers are intentionally overlapping. Each layer has a different physical or logical failure mechanism and is designed to be independently reviewable and testable.

Layer	Function	Hardware / Implementation
1	Occupant Survival Capsule	Central fixed capsule with thick multi-angle carbon-fiber roll-cage connecting body panels from all sides
2	Multi-stage Energy Absorption	Landing-leg response → airframe controlled deformation → capsule-mount stroke → roll-cage → soft damper seats
3	Capsule-to-Airframe Isolation	Multiple high-stroke mechanical damper mounts between capsule and main structure
4	Distributed Propulsion	7 independent rotor channels; loss of one channel does not remove all lift authority
5	Prime-Mover Redundancy	4 independent turboshaft units; any single unit can be isolated
6	Fire Detection / Isolation / Suppression	Zone detection, fuel/electrical isolation, suppression around central rotor and capsule perimeter
7	Control Redundancy	Segregated sensing, computation and actuator paths with explicit fault-containment logic
8	Emergency Descent / Recovery	8 advanced parachutes (aircraft-level) + controlled degraded flight modes
9	Water Survival	Buoyancy provisions, flotation concept, multi-orientation egress
10	Structural Containment & Modular Jettison	High-energy hazards kept outside occupant volume; modular isolation/jettison of a burning rotor after isolation

Safety claim discipline: no layer is treated as a guaranteed survival device at this stage. The safety case must quantify failure probabilities, load cases, deployment envelopes, occupant injury criteria, and test evidence before certification credit is claimed. No under-capsule fans are included in this fixed-capsule baseline. Emergency descent provisions remain aircraft-level and subject to validation.

6. Fixed Safety Capsule – Detailed Features

The flagship 10-ton version uses a fixed, non-jettisonable safety capsule. The capsule remains mechanically integrated with the aircraft but is intentionally isolated from direct structural shock paths. This removes separation dynamics, independent emergency propulsion sizing, and complex synchronization from the first aircraft while still providing a highly protected occupant volume.

Feature	Specification
Primary volume shape	Near-spherical / highly rounded for multi-direction impact resistance
Internal reinforcement	Thick multi-angle carbon-fiber roll-cage connecting body panels from all sides
Occupant seats	7 very soft bed-like seats with high-stroke mechanical dampers to the capsule floor
Capsule-to-airframe interface	Multiple high-stroke mechanical damper mounts
Capsule landing legs	6 mechanical damper legs with progressive energy absorption
Main aircraft landing legs	7 mechanical damper legs with thick soft plastic tip sections
Fire protection around capsule	Suppression nozzles at perimeter
Fire protection around central rotor	Dedicated detection and suppression zone
Egress concept	Multiple exit routes; operable after normal electrical power loss (design objective)
Parachute system	8 advanced canopies sized for whole-aircraft emergency recovery (concept)

Conceptual energy-absorption sequence: (A) Landing-leg and outer structure response → (B) Primary airframe controlled deformation → (C) Capsule support stroke → (D) Capsule shell and roll-cage load distribution → (E) Seat and restraint attenuation. The capsule should not be described as “crash-proof.” The engineering target is a defined survivability envelope.

7. Seven-Rotor Propulsion Architecture (6.0 m Diameter)

The seven primary rotors are the central lift system. Each rotor has a nominal 6.0 m diameter. The preliminary geometric disk area is approximately 197.9 m² for seven rotors combined. One rotor is located on the central roof line; six rotors are arranged in a surrounding ring.

Item	Preliminary value
Rotor count	7
Rotor diameter	6.0 m
Radius	3.0 m
Area per rotor	28.27 m²
Total geometric disk area	197.9 m²
Disk loading at 10,000 kg	~50.5 kg/m²
Disk loading at 15,000 kg lift case	~75.8 kg/m²
Hover thrust per rotor at 10 t	~14.01 kN
Illustrative 20% total-thrust margin point	~16.81 kN per rotor
Ideal induced hover power (10 t, ISA SL)	~1.40 MW
Approximate real power (FM ≈ 0.72)	~1.94 MW

The increase from 5.6 m to 6.0 m diameter improves disk loading and reduces the power required for hover, providing a more comfortable design margin for the 10-ton class while still supporting a conceptual 15-ton maximum lift case. Real installed power must include induced, profile, mechanical, transmission, electrical, control, atmospheric, and reserve margins.

Each rotor is conceived as a modular unit with its own structural interface, drive system, monitoring and isolation capability. Local vibration, temperature, electrical and fire monitoring, safe isolation of power/fuel/fluid/data interfaces before any emergency release, and prevention of failure propagation to adjacent modules are mandatory design objectives.

8. Propulsion and Engine Architecture (Closed Direction)

This revision closes the previously open propulsion direction. Four turboshaft-class prime movers are selected as the baseline architecture. They feed a segmented high-voltage electrical distribution system that supplies seven independent rotor drive channels. This arrangement allows any single prime mover or any single rotor channel to be isolated without immediate total loss of lift authority.

Item	Target / Concept
Number of prime movers	4
Power class per unit	Approximately 650–700 kW class (final OEM selection pending)
Total installed power target	2.5 – 2.8 MW
Analytical rationale	Real hover power estimate ~1.94 MW at 10 t; installed target provides ~25–40% system margin
Benchmark engine class	PT6C-family or equivalent turboshaft in the 650–1,000+ kW published class (installation derate applies)
Power flow	Turboshaft → generator → segmented HV bus → independent inverters → rotor motors
Key safety rule	Redundancy must be physically independent; shared vulnerable buses do not equal independent channels
Degraded operation objective	Continue controlled flight / landing after loss of any single prime mover or any single rotor channel

Final engine selection, generator matching, installation rights, export controls and certification pathway require direct OEM engagement. Published engine figures are engine-level only; installed aircraft power after generators, conversion losses, thermal limits and accessories will be lower. Alternative candidates remain open for trade studies once a frozen shaft/electrical power curve exists.

9. Fuel System, Endurance and Range

Fuel capacity is set at 1,700 L nominal. Fuel is treated as a high-consequence energy source and participates directly in the safety architecture: segregated containment away from the primary occupant capsule, fuel shutoff capability following crash/fire/severe fault, fire-resistant routing, leak detection, controlled venting, and thermal shielding between hot machinery and fuel compartments.

Parameter	Value
Nominal tank capacity	1,700 L
Reference fuel mass	~1,360 kg (at 0.80 kg/L Jet-A)
Approximate real hover power at 10 t	~1.94 MW
Hover time at 10 t (SFC ~0.32–0.35 kg/kWh, typical)	Approximately 115–130 minutes
Cruise endurance (typical mission profile)	Approximately 3.2–3.6 hours
Approximate range at 200 km/h (with normal reserves)	Approximately 620–720 km

These endurance and range figures are preliminary order-of-magnitude estimates based on the closed power direction and 1,700 L capacity. They depend on actual installed efficiency, mission profile, reserves policy and atmospheric conditions. They are not certified performance guarantees. Early flight tests should use reduced fuel mass wherever the test objective allows it.

10. Preliminary Mass Budget

The following is a planning-level mass allocation intended to keep the 10-ton baseline internally visible. It is not a certified weight statement. The allocation must be replaced by component-level supplier data as the design matures. The larger 6.0 m rotors and 1,700 L fuel are reflected.

Mass group	Planning mass	Comment
Primary airframe and rotor support structure	2,050 kg	Includes major load paths and rotor support
Safety capsule and occupant protection structure	1,150 kg	Shell, roll-cage, isolators, dampers, seats
Seven rotors, hubs and pitch/control hardware (6.0 m)	720 kg	Updated for 6.0 m; supplier-dependent
Powerplant / generators / motors / drives (4 units)	1,550 kg	Four turboshaft-class + electrical chain
Fuel system excluding fuel	230 kg	Tanks, pumps, plumbing, protection
Fuel, 1,700 L planning capacity	1,360 kg	Planning density 0.80 kg/L
Avionics, flight control and electrical distribution	520 kg	Includes redundancy allowance
Landing gear / ground interface (7+6 dampers)	430 kg	Main + capsule mechanical legs
Fire, emergency recovery and safety equipment	540 kg	Includes 8-parachute concept allocation
Thermal management / cooling	350 kg	Powerplant-dependent
Interconnects, hardware, fluids and growth reserve	1,200 kg	Development margin
Total planning mass	≈ 9,800 kg	Leaves ~200 kg to the 10,000 kg target

The residual is intentionally not assigned to a passenger or payload claim. It is a design-closure reserve until the propulsion and structure selections are validated with supplier data. Continuous bottom-up mass-statement discipline is mandatory.

11. Flight-Control and Fault-Reconfiguration Architecture

The seven-rotor layout is intended to support distributed control and graceful degradation. The flight-control system must detect abnormal rotor speed, thrust, temperature, vibration, power, and actuator behavior and then reconfigure the control law within validated limits.

- Independent monitoring of each rotor channel.
- Cross-checking of commanded versus measured rotor performance.
- Automatic isolation of a failed propulsion channel where appropriate.
- Reallocation of collective and attitude authority across healthy rotors.
- Pilot/vehicle warning hierarchy that distinguishes degraded operation from immediate emergency.
- Independent emergency power for flight-critical control functions.
- Hardware-in-the-loop and fault-injection testing before flight.

A critical design principle is that fault reconfiguration must not create a new unstable condition. Every claimed failure case therefore requires a defined control-law state, actuator authority assessment, structural load assessment, and flight-test envelope.

12. Structural and Load-Path Philosophy

The primary structure is organized into multiple load paths around the central capsule. Rotor loads, engine loads, landing loads, and emergency loads are not assumed to follow the same path.

- Rotor stations have dedicated local load paths into the primary frame.
- Capsule loads are transferred through controlled interfaces and dampers.
- Landing loads are routed through landing structures and energy absorbers before reaching the capsule.
- High-consequence joints receive dedicated inspection and test requirements.
- Structural sizing will use limit and ultimate load cases and must account for load redistribution.

EASA's CS-27 framework states that rotorcraft strength requirements are expressed in limit and ultimate loads and, unless otherwise provided, uses a 1.5 factor of safety. The eventual certification basis will depend on the aircraft category and certification authority selected for the product. This document does not claim compliance. It uses certification principles as design-development guidance only.

13. Emergency Recovery Concept

Emergency recovery is intentionally treated as a layered function rather than a single parachute or a single capsule mechanism.

Scenario	Primary response	Secondary response	Validation required
Single rotor failure	Distributed control reconfiguration	Power redistribution / landing mode	Simulation + flight test
Multiple rotor degradation	Emergency flight mode	Aircraft recovery system (8 parachutes)	Failure-injection analysis + test
Major fire	Detection, isolation, power reduction	Emergency recovery / landing	Fire tests + system tests
Severe structural event	Maintain capsule integrity	Aircraft recovery concept	Dynamic structural test
Water impact / ditching	Capsule survivability and egress	Recovery / flotation provisions	Water-impact test

The emergency recovery concept includes an aircraft-level parachute system with a baseline concept of eight advanced canopies. Exact canopy count, deployment altitude, reefing strategy, attachment loads, and certification credit are not frozen by this revision and remain validation items.

14. Fire and Thermal Safety

- Thermal sensors in engine, generator, fuel and high-energy zones.
- Fire detection with independent logic from propulsion control where practical.
- Fuel isolation valves and emergency shutoff.
- Fire-resistant barriers between hot machinery and occupied space.
- Suppression system sized after hazard analysis (perimeter of capsule + central rotor zone).
- Emergency power reduction and load shedding.
- Post-fire safe-state logic.

The safety philosophy is based on containment plus isolation. The goal is not to assume that a fire can always be extinguished instantly; it is to prevent a local event from propagating into a loss of the occupied capsule.

15. Water Survival and Ditching

The capsule is intended to provide a protected survival volume during a water-landing scenario. The concept includes structural protection, controlled flotation, protected egress, and preservation of critical emergency equipment.

- Watertight or controlled-leakage boundaries around the occupant volume.
- Flotation provisions sized after mass-property and stability analysis.
- Emergency egress accessible after defined attitudes.
- Independent emergency electrical power.
- Location and activation logic for survival equipment.
- Capsule-to-airframe interfaces designed to limit water-impact shock transmission.

Water-impact survivability is a major validation item. It cannot be established from conceptual geometry alone.

16. Manufacturing and Prototyping Strategy

The compact baseline is deliberately structured for staged physical development. The recommended development path is not to build the complete aircraft immediately.

- Phase 1: coupon and material tests for structural and energy-absorption elements.
- Phase 2: subscale rotor, motor, controller and thermal-management tests.
- Phase 3: full-scale capsule impact and damper test articles.
- Phase 4: integrated propulsion test rig with multiple rotor channels and four prime-mover simulation.
- Phase 5: unmanned integrated demonstrator.
- Phase 6: progressive flight-envelope expansion.
- Phase 7: certification-oriented conformity and compliance programme.

This staged approach is a major reason for selecting the 10-ton baseline. The demonstrator can validate the safety architecture without requiring the resources of the superseded very-large aircraft concept.

17. Verification and Validation Plan

Subsystem	Analysis	Bench test	Dynamic test	Flight test
Rotor / hub	Yes	Yes	Yes	Yes
Propulsion channel	Yes	Yes	Yes	Yes
Flight controls	Yes	HIL	Fault injection	Yes
Capsule structure	FEA	Component	Full-scale impact	Limited / integrated
Dampers	Model	Characterization	Full-scale	Integrated
Fire protection	Hazard analysis	Component	System fire test	As required
Emergency recovery	Simulation	Component	Deployment test	Flight demonstration
Water survival	Simulation	Tank tests	Water-impact test	As required

The verification programme should trace every safety requirement to a measurable verification method and evidence record. No investor-facing performance claim should be upgraded from target to demonstrated performance without test evidence.

18. Certification Strategy

The certification pathway is not fixed in Document 00. The aircraft's final category, operational concept, occupant capacity, maximum takeoff mass, and jurisdiction will determine the applicable certification basis.

EASA currently maintains CS-27 for small rotorcraft and CS-29 for large rotorcraft. Current EASA material also emphasizes structural loads, critical parts, and test substantiation. For a new rotorcraft, structural substantiation may require analysis supported by component, subcomponent, or full-scale testing.

The project should therefore maintain a certification-assumption register and avoid presenting this conceptual document as evidence of compliance.

19. Safety Assessment Framework

- Functional Hazard Assessment for flight-critical and occupant-protection functions.
- Preliminary System Safety Assessment for propulsion, flight control, fire protection, emergency recovery and capsule systems.
- Common-cause and common-mode failure analysis.
- Failure propagation analysis between fuel, hot machinery, electrical systems and capsule.
- Structural failure and load-path assessment.
- Human factors and emergency egress assessment.
- Fault detection and annunciation assessment.
- Verification traceability from requirement to test evidence.

The ten-layer architecture is not ten independent guarantees. Several layers can share common causes of failure. The safety case must therefore explicitly analyze common-mode vulnerabilities and dependencies.

20. Product and IP Positioning

The compact 10-ton platform is intended to become the primary demonstrator and commercial flagship baseline. The strongest differentiating theme is not merely 'a helicopter with a capsule'; it is an integrated architecture in which the capsule, structural isolation, energy absorption, distributed propulsion, fire containment, and emergency recovery are designed as one safety system.

- Core architecture: fixed safety capsule + 7 × 6.0 m distributed rotor system.
- Mechanical isolation and energy-absorption architecture.
- Fault-tolerant distributed propulsion/control architecture (4 prime movers + 7 rotor channels).
- Fire and energy containment architecture.
- Emergency recovery and survival architecture (8-parachute concept).
- Modular interfaces supporting future detachable-capsule derivatives.

Any patent strategy should be developed from verified claimable mechanisms and carefully separated from generic aerospace features. This document is a technical baseline, not a legal opinion on patentability or freedom-to-operate.

21. Preliminary Risk Register

Risk	Impact	Current treatment	Next closure action
Mass growth above 10 t	High	Growth reserve retained	Supplier quotes + weighed hardware
Installed power higher than target	High	2.5–2.8 MW target set with margin	Rotor CFD + system efficiency model
Damper stroke insufficient	High	Multiple layers specified	Dynamic impact testing
Capsule structure too heavy	Medium/High	Dedicated mass allocation	FEA + material coupons
Common-mode control failure	High	Distributed architecture	SSA + HIL fault injection
Fire propagation	High	Containment + isolation layers	Fire hazard analysis + tests
Emergency recovery load exceedance	High	8-parachute concept retained	Deployment simulation + structural test
Water-impact instability	High	Capsule survival concept	Water-impact modelling + test
Thermal management mass growth	Medium	Dedicated allocation	Thermal model + rig test
Engine/generator integration mismatch	High	Early OEM engagement required	Matching studies + supplier data

22. Open Engineering Items Before Rev 01

- Freeze detailed mission profile: payload, crew, cruise speed, altitude, range and endurance.
- Complete rotor aerodynamic model and define rotor RPM, blade geometry, solidity, tip speed and control method for the 6.0 m rotors.
- Final OEM selection and generator matching for the four turboshaft-class units.
- Close full power budget including realistic hover, climb and degraded-mode margins.
- Develop first-order transmission, generator, motor and inverter thermal model.
- Replace planning mass allocations with component-level estimates and supplier data.
- Develop capsule structural load cases and damper stroke/force targets.
- Define exact parachute sizing, deployment envelope and structural attachment loads.
- Define water-impact and flotation requirements in quantitative terms.
- Create a formal system safety architecture and failure-mode database.
- Map the final design to the applicable certification basis.

Rev 01 should be triggered by closure of the detailed rotor aerodynamic model, OEM engine/generator package, first structural sizing loop, first capsule energy-absorption model, and a controlled component-level mass budget.

23. Design Rules for the 10-Ton Baseline

Rule 1

Do not allow the gross-mass target to become a paper-only number. Every major subsystem must carry a measurable mass allocation.

Rule 2

Do not claim safety credit for an emergency device before its deployment, structural, and human-survivability evidence exists.

Rule 3

Prefer passive mechanical safety layers where they can reduce dependence on software or electrical power.

Rule 4

Avoid single common-mode dependencies that can defeat multiple safety layers simultaneously.

Rule 5

Treat the capsule as a system, not simply as a stronger cabin.

Rule 6

Design interfaces for future modular derivatives without forcing the first demonstrator to carry the full complexity of the detachable version.

Rule 7

Maintain explicit separation between controlled baseline values, engineering targets, calculated estimates, and demonstrated performance.

24. Preliminary Performance Envelope

Document 00 presents the following performance-related values as conceptual estimates only. They depend strongly on the unresolved detailed aerodynamic, rotor, and installed-efficiency models.

Metric	Current treatment
Gross mass	10,000 kg target
Hover thrust requirement	~98.1 kN total; ~14.01 kN per rotor
Illustrative 20% total thrust margin	~117.7 kN total; ~16.81 kN per rotor
Ideal induced hover power floor	~1.40 MW at ISA sea-level density
Approximate real hover power (FM ≈ 0.72)	~1.94 MW
Installed power target	2.5 – 2.8 MW
Hover endurance (1,700 L, typical SFC)	Approximately 115–130 minutes
Cruise endurance (typical)	Approximately 3.2–3.6 hours
Approximate range at 200 km/h	Approximately 620–720 km (with reserves)
Cruise speed	TBD after detailed aerodynamic model

The ideal and approximate real hover figures are sanity checks only. They are not claims of achievable certified aircraft performance.

25. Investor and Due-Diligence Readiness

The 10-ton baseline with closed propulsion direction (4 turboshaft-class units, 2.5–2.8 MW target), 6.0 m rotors, 1,700 L fuel and ten-layer safety architecture is intentionally easier to communicate than the superseded very-large platform. The commercial story can be built around a demonstrable safety architecture, a compact physical demonstrator, and a staged validation programme.

- Clear controlled baseline with frozen rotor geometry and propulsion direction.
- Visible safety architecture rather than a single headline feature.
- Prototype-friendly scale.
- Modular propulsion and control interfaces.
- Traceable engineering risks.
- Explicit distinction between concept, calculated estimate, and demonstrated result.

A professional investor package should pair this document with a one-page product sheet, a mass/power spreadsheet, a preliminary hazard analysis, a test roadmap, and evidence from physical demonstrators. The existence of a polished document alone should never be represented as proof of flight performance or certification.

26. Glossary

Term	Meaning
CDD	Conceptual Design Document.
T/W	Thrust-to-weight ratio.
HIL	Hardware-in-the-loop.
FEA	Finite element analysis.
CFD	Computational fluid dynamics.
LRU	Line-replaceable unit.
MTOM	Maximum take-off mass.
FHA	Functional Hazard Assessment.
SSA	System Safety Assessment.
Fixed capsule	Safety capsule that remains mechanically integrated with the aircraft and is not designed to separate in the baseline configuration.
Graceful degradation	Controlled reduction in capability following defined failures while preserving required control or landing functions.
Installed power	Total shaft/electrical power available at the aircraft level after installation losses.
FM	Figure of Merit (rotor efficiency metric).

27. Final Baseline Statement

Document 00 (Revision 00-T10-R6-Full) establishes the complete Gen-3 compact flagship baseline: a 10-ton, seven-rotor rotorcraft with seven nominal **6.0 m** rotors (total disk area $\approx 197.9 \text{ m}^2$), **1,700 L** fuel capacity, four turboshaft-class prime movers with an installed power target of **2.5–2.8 MW**, a $17 \text{ m} \times 17 \text{ m} \times 4 \text{ m}$ target envelope, and a fixed non-jettisonable safety capsule protected by a ten-layer safety architecture.

The central engineering proposition is layered survivability. The aircraft does not depend on one rescue mechanism. It combines structural containment, mechanical isolation, multiple damped load paths, fire protection, distributed propulsion with four independent prime movers and seven independent rotor channels, fault reconfiguration, emergency recovery concepts (including an 8-parachute aircraft-level system), and multi-stage occupant energy management.

The baseline remains deliberately conservative about claims. Quantitative performance, mass, power, recovery performance, and certification compliance remain subject to analysis and test. The next revision should close the detailed rotor aerodynamic model, OEM engine/generator package, and first component-level mass statement, because those disciplines drive the feasibility of nearly every downstream system.

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